Activity 8

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Identify seasonality patterns

1) Series plot & Trend

No of observation n=70

Cycle length = 7 days

Full cycles: 10

Overall trend: Rising

Slope: 81.32 riders per day.

Ridership increases over time with a weekly repeating pattern.

2) Seasonal pattern:-

Peak day: Day 5, Index ≈ 1.281

Trough day: Day 2, Index ≈ 0.729

Seasonal swing = 1.281 − 0.729 = 0.552

The seasonal pattern repeats every 7 days.

3) Seasonal consistency

* Peak occurs at day 5 in 10 out of 10 cycles
* The seasonal timing is consistent.
* The size of the seasonal swings increase

with ridership.

4) Autocorrelation (ACF)

* Significance bound :- ±0.239
* ACF at lag 3 :- −0.839
* ACF at lag 7 : 0.869
* Lag 7 shows strong positive autocorrelation, confirming weekly seasonality

5) Seasonality type and conclusion

* First cycle range :- 3099 riders
* Last cycle range :- 5569 riders
* Range ratio. 5569/3099 ≈ 1.80
* Chosen type :- multiplicative seasonality.
* Adjust trend slope approximately 80 riders per day.

management advice :- Increase bus availability and staffing around days, and prepare for growing passenger demand over time.